
How Much Noise Can Adaptive-Variational Berry-Phase Computation Tolerate?

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Abstract

The Berry phase is a geometric observable that distinguishes topologically trivial from nontrivial phases of matter. Adaptive variational quantum dynamics (AVQDS) computes it in circuits that are orders of magnitude shallower than Trotterized adiabatic evolution, but the published benchmarks are noiseless state-vector simulations. We characterize the resilience of this algorithm to realistic hardware noise, asking how large the incoherent error can be before the computed Berry phase degrades. We re-implement the AVQITE+AVQDS pipeline on the four-site Su–Schrieffer–Heeger–Hubbard chain in both the noninteracting ($U = 0$) and strongly correlated ($U = 10$) regimes, reproduce the noiseless $\varphi_B \in \{0, \pi\}$, and then re-evaluate the full measured phase $\varphi_B = \varphi_{qc} + \varphi_G$ —both the Hadamard-test readout and the geometric-phase integrands—under a genuine density-matrix noise model with finite-shot sampling, two-qubit depolarizing error, ancilla readout error, and T_1/T_2 decoherence, on a frozen noiseless trajectory. We extract, for each regime, the largest incoherent-noise strength at which the estimate stays within a pre-stated tolerance, report it in device-referenced units, and explain why the threshold sits where it does. We find that the noiseless robustness of the method—the cancellation of non-adiabatic residuals between the circuit phase and the geometric phase—does not fully survive stochastic noise: the deterministic bias the two legs share partially cancels, but their independent fluctuations add, and the dynamical-phase leg accumulates a non-cancelling drift. The interacting regime is the more fragile of the two by roughly two orders of magnitude in the noise strength: its 3–4 $\times$ deeper circuit amplifies the incoherent gate error through the interaction energy scale and the large parameter velocities of the over-parameterized ansatz, pushing the threshold from $\sim 10^{-1}$ ($U=0$) down to $\sim 10^{-3}$ ($U=10$).

1 Introduction

The Berry phase [1] underlies the quantum Hall effect, topological insulators, and topological superconductors. For a one-dimensional time-reversal-invariant crystal it is quantized to 0 (trivial) or π (nontrivial), and computing it for interacting many-body systems is classically hard, which motivates quantum algorithms. A recent work by Mootz and Yao [2] combines adaptive variational quantum imaginary-time evolution (AVQITE) for ground-state preparation with adaptive variational quantum dynamics simulation (AVQDS) for cyclic adiabatic evolution, and reads the accumulated phase off a Hadamard-test circuit. The advance is efficiency: the adaptive ansatz produces circuits a few tens to a few hundred layers deep, orders of magnitude shallower than a Trotterized adiabatic loop at matched accuracy.

The reported results are, however, entirely noiseless: “neglecting errors from quantum gates and finite sampling.” The question this leaves open is the one that decides whether the shallow-circuit advantage matters in practice: under realistic device noise, how accurate does the Berry-phase estimate remain, and at what noise level does it break?

This is not a routine parameter sweep, for two reasons. First, the Berry phase is read out from a measured probability, so finite-shot sampling noise is intrinsic and irreducible: it persists even on a fault-tolerant device, because the final logical measurement is still a projective sample. Second, the noiseless robustness of the method—it returns the correct phase even under strong non-adiabaticity—comes from two phase contributions that co-evolve along the same variational trajectory so that their non-adiabatic errors cancel, $\varphi_B = \varphi_{qc} + \varphi_G$. That cancellation is a property of the co-evolving, adaptively grown ansatz, and it does not automatically carry over to stochastic device noise: the two contributions are sampled independently, so their fluctuations add.

In this work we (i) re-implement the AVQITE+AVQDS Berry-phase pipeline and reproduce the noiseless anchor, (ii) state a complete noise model, phase estimator, accuracy criterion, and threshold-extraction rule, (iii) re-evaluate the whole measured phase under density-matrix noise at the frozen noiseless trajectory, and (iv) extract a device-referenced noise threshold together with a mechanism analysis of why it sits where it does.

2 Method

2.1 The algorithm under test

We hold the algorithm fixed as in the original paper. A Hamiltonian $\hat{\mathcal{H}}(\lambda)$ depends on a cyclic parameter $\lambda \in [0, 2\pi]$ with $\hat{\mathcal{H}}(2\pi) = \hat{\mathcal{H}}(0)$; the ground state $|G(\lambda)\rangle$ accumulates a dynamical phase φ_D and a Berry phase $\varphi_B = -i \oint d\lambda \langle G(\lambda) | \partial_\lambda G(\lambda) \rangle$ under one cycle. A time-reversal-symmetric loop $U_{\text{loop}} = U_\odot(0, T/2) U_\odot(T/2, 0)$ cancels φ_D while φ_B adds.

The variational ansatz is $|\Psi[\theta]\rangle = \prod_\mu e^{-i\theta_\mu \hat{A}_\mu} |\Psi_0\rangle$. AVQITE grows it to prepare $|G(0)\rangle$; AVQDS grows it further to track the cyclic evolution, keeping the McLachlan distance below $L_{\text{cut}}^2 = 10^{-4}$ and integrating the equation of motion $\sum_\nu M_{\mu\nu} \dot{\theta}_\nu = V_\mu$ with RK4 at $\delta\theta_{\text{max}} = 0.01$. The Berry phase is

$$\varphi_B = \varphi_{qc} + \varphi_G, \quad \varphi_G = \varphi_{G,1} + \varphi_{G,2}, \quad (1)$$

with $\varphi_{G,1} = -\int \langle \hat{\mathcal{H}} \rangle_\theta dt$ (the dynamical leg) and $\varphi_{G,2} = \int \text{Im} [\sum_\mu \langle \partial_{\theta_\mu} \Psi | \Psi \rangle \dot{\theta}_\mu] dt$ (the geometric leg). φ_{qc} is read from the Hadamard test on the overlap $\langle \Psi[\theta(0)] | \Psi[\theta(2\pi)] \rangle$.

2.2 Test bed and noiseless anchor

The test bed is the four-site dimerized Fermi–Hubbard (Su–Schrieffer–Heeger–Hubbard) chain, $t = 1$, mapped to eight qubits by Jordan–Wigner, with the twist applied to spin-up fermions only. We study $U = 0$ and $U = 10$, each in a topologically trivial ($\delta > 0$, $\varphi_B^{\text{ED}} = 0$) and nontrivial ($\delta < 0$, $\varphi_B^{\text{ED}} = \pi$) dimerization, at loop time $T = 20$. The exact-diagonalization references are computed at runtime by our own Wilson-loop pipeline (logged in `proposal/logs/anchor_*.json`) and return π ($\delta < 0$) and 0 ($\delta > 0$) in all four configurations.

Our noiseless re-implementation reproduces the quantized phases to within 4×10^{-3} rad in all four configurations (Table 1). The circuit resources match the paper’s order: 18–25 layers and 148–260 CNOTs for $U = 0$, versus 70–98 layers and 690–910 CNOTs for $U = 10$ (the interacting circuits are 3–4× deeper). In the nontrivial $U = 0$ regime the circuit is the larger one within that pair, as the paper reports; for $U = 10$ the *trivial* loop is the larger circuit (98 layers, 910 CNOTs versus 70 layers, 690 CNOTs), because AVQDS grows more operators to track the time-reversed trivial loop at the fixed L^2 cutoff.

2.3 Noise model

We model four incoherent sources, applied jointly and scaled by a single dimensionless strength ξ against a fixed base device profile:

- *two-qubit depolarizing* after every CNOT, rate $p_2 = \xi p_2^{(0)}$ with $p_2^{(0)} = 10^{-3}$;
- *single-qubit depolarizing* after every one-qubit gate, rate $p_1 = \xi p_1^{(0)}$ with $p_1^{(0)} = 10^{-4}$;

- *ancilla/system readout error*, a symmetric bit-flip with probability $p_{\text{ro}} = \xi p_{\text{ro}}^{(0)}$ with $p_{\text{ro}}^{(0)} = 10^{-2}$;
- T_1/T_2 *decoherence*, amplitude plus phase damping keyed to gate times $t_{1q} = 30$ ns and $t_{2q} = 300$ ns with $T_1^{(0)} = 100$ μ s and $T_2^{(0)} = 70$ μ s, scaled so the damping rates are $\propto \xi$.

The four channels co-vary along ξ in the ratio fixed by this profile, so a threshold quoted at a reference shot count S_{ref} is interpretable as “accurate below roughly X two-qubit error at S_{ref} shots.” Finite-shot sampling is applied to every measurement.

Noise is injected on the actual circuit in a genuine density-matrix simulation: each Pauli rotation is compiled to basis-change gates, a CNOT ladder, an $R_z(2\theta)$ rotation, and the uncompute, and the channels act after each gate. In this study the ansatz and the variational trajectory $\theta(t)$ are frozen at their noiseless values (the McLachlan integration is not re-run under noise), but the quantities the device must measure to evaluate φ_G —the energy expectation $\langle \hat{\mathcal{H}} \rangle$ feeding $\varphi_{G,1}$ and the geometric-connection overlaps $\langle \Psi | \partial_{\theta_\mu} \Psi \rangle$ feeding $\varphi_{G,2}$ —are re-evaluated under the same noise as the readout, so both φ_{qc} and φ_G carry stochastic noise and the extracted threshold is a whole-method number. In $\varphi_{G,2}$ the frozen noiseless $\dot{\theta}_\mu(t)$ multiplies the noisy connection overlaps.

2.4 Phase estimator and branch policy

The single real-part Hadamard test is ill-conditioned wherever φ_{qc} sits near 0 or π , because $P_{|0\rangle} = \frac{1}{2}(1 + \cos \varphi_{\text{qc}})$ has vanishing slope there. We therefore measure both the real and the imaginary parts of the overlap (the imaginary part via a $S^\dagger$ gate before the final Hadamard) and estimate $\varphi_{\text{qc}} = \text{atan2}(\text{Im}, \text{Re})$, which is well-conditioned across the loop. The geometric connection is measured as $\langle \partial_{\theta_\mu} \Psi | \Psi \rangle = i \langle \mathcal{A}_\mu \rangle_{\phi_\mu}$ on the sub-circuit state $|\phi_\mu\rangle = U_{\mu-1} \cdots U_0 |\Psi_0\rangle$, i.e. a Pauli expectation on a device-preparable state.

Because the raw phase for $U = 10$ comes out as an odd multiple of π , all deviations $|\varphi_B - \varphi_B^{\text{ED}}|$ and the trivial/nontrivial discrimination margin are computed *on the circle*: estimate and reference are mapped to the same principal value and the shorter angular distance is taken.

2.5 Accuracy criterion and threshold rule

We declare the estimate accurate when its mean deviation (over seeds, on the circle) from the exact-diagonalization reference is below $\delta_{\text{tol}} = 0.15$ rad. The noise threshold ξ^* is the largest grid value ξ for which the mean deviation remains below δ_{tol} ; its uncertainty is reported from the seed spread and the grid spacing. This rule is fixed before any results are inspected.

3 Results

3.1 Noise-resilience curves

Figure 1 shows the Berry-phase deviation versus the noise strength ξ at $S = 10^4$ shots, for all four configurations, with seed error bars (three seeds per point). For $U = 0$ the deviation starts from a finite-shot floor of a few times 10^{-2} rad at low noise, crosses the 0.15 rad tolerance between $\xi = 0.03$ and $\xi = 0.3$ (nontrivial) or between $\xi = 0.03$ and $\xi = 0.1$ (trivial), and eventually saturates near the value expected for a uniformly random phase. For $U = 10$ the two curves are displaced ~ 30 – $100\times$ to smaller ξ : the deviation crosses the tolerance between $\xi = 0.003$ and $\xi = 0.01$ (nontrivial) and between $\xi = 0.001$ and $\xi = 0.003$ (trivial), then rises steeply and saturates, reflecting the larger gate error that the deeper interacting circuit accumulates at the same ξ .

3.2 Thresholds

Table 1 reports, for each configuration, the noiseless and under-noise Berry phase, the deviation, the extracted threshold in device-referenced units, and the circuit resources. The threshold is reported separately for the shallow ($U = 0$) and deep ($U = 10$) circuits so the depth dependence is visible; where the trivial and nontrivial regimes of the same U give different thresholds, the per- U threshold

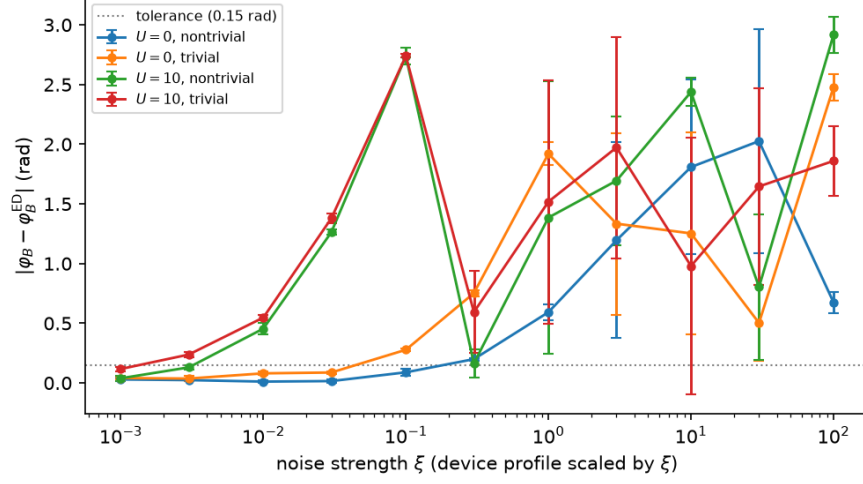


Figure 1: Berry-phase deviation versus noise strength ξ (device profile scaled by ξ , $S = 10^4$ shots), with seed error bars.

Table 1: Main results. $|\Delta\varphi_B| = |\varphi_B - \varphi_B^{\text{ED}}|$ is the angular deviation on the circle; the threshold is in device-referenced units (an equivalent two-qubit error p_2 , readout error p_{ro} , and T_1/T_2 at the base profile), all quoted at $S_{\text{ref}} = 10^4$ shots.

Configuration	Noise setting	φ_B	$ \Delta\varphi_B $	ξ^*	Depth	N_{CX}
$U=0$, nontrivial	noiseless	π	0.0040	$0.1^{+0.2}_{-0.07}$	25	260
$U=0$, nontrivial	at threshold	0.97π	0.089 ± 0.030	$0.1^{+0.2}_{-0.07}$	25	260
$U=0$, trivial	noiseless	0	0.0033	$0.03^{+0.07}_{-0.02}$	18	148
$U=0$, trivial	at threshold	-0.03π	0.087 ± 0.011	$0.03^{+0.07}_{-0.02}$	18	148
$U=10$, nontrivial	noiseless	π	0.0029	$0.003^{+0.007}_{-0.002}$	70	690
$U=10$, nontrivial	at threshold	-0.96π	0.130 ± 0.019	$0.003^{+0.007}_{-0.002}$	70	690
$U=10$, trivial	noiseless	0	0.0004	$0.001^{+0.002}_{-0.001}$	98	910
$U=10$, trivial	at threshold	0.04π	0.117 ± 0.020	$0.001^{+0.002}_{-0.001}$	98	910

below which the two topological phases stay distinguishable is the more conservative (smaller) of the two.

For $U = 0$ the nontrivial threshold is $\xi^* = 0.1$ and the trivial $\xi^* = 0.03$, so the conservative per- U threshold is $\xi^* = 0.03$, i.e. an equivalent two-qubit error $p_2 \approx 3 \times 10^{-5}$, readout error $p_{\text{ro}} \approx 3 \times 10^{-4}$, and coherence times T_1/T_2 roughly $1/\xi^* \approx 30\times$ longer than the base profile, at $S_{\text{ref}} = 10^4$ shots. The minimum-shots floor is $S^* \approx 1000$ shots.

For $U = 10$ the threshold is strikingly lower: the deviation crosses the 0.15 rad tolerance between $\xi = 0.003$ and $\xi = 0.01$ (nontrivial) and between $\xi = 0.001$ and $\xi = 0.003$ (trivial), so $\xi^* = 0.003$ (nontrivial) and $\xi^* = 0.001$ (trivial), i.e. an equivalent two-qubit error near 10^{-6} and readout error near 10^{-5} at $S_{\text{ref}} = 10^4$ shots. This is a gate-noise effect, not a shot-count effect: re-running the $U = 10$ sweep at $S_{\text{ref}} = 3 \times 10^5$ shots leaves the deviation curve unchanged to within the seed spread, so the threshold is set by the incoherent error that the deep circuit accumulates, not by the finite sampling. The depth/energy-scale dependence is clean: the interacting circuit is $3\text{--}4\times$ deeper and its energy expectation carries coefficients of size ~ 10 , and together these lower the threshold by roughly two orders of magnitude relative to $U = 0$.

3.3 Shot-noise scaling

Because the Berry phase is read from a measured probability, the finite-shot sampling noise is irreducible and must be characterized on its own axis. Figure 2 shows the seed standard deviation of the deviation versus the shot count S at fixed noise ($\xi = 0.03$). The uncertainty falls as $\sim S^{-1/2}$ in

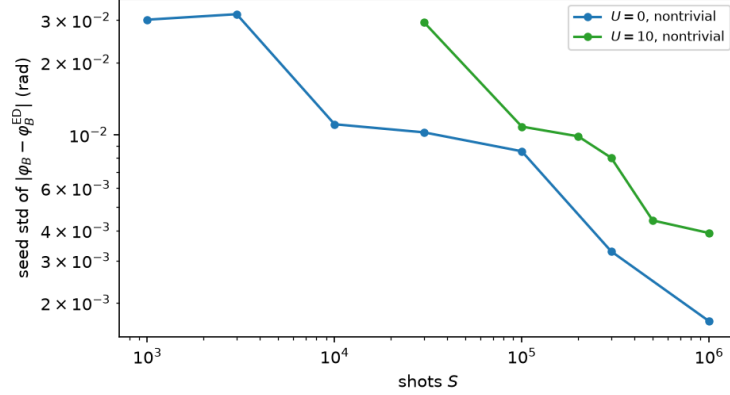


Figure 2: Seed standard deviation of the Berry-phase deviation versus shot count S at fixed $\xi = 0.03$. The dashed guide shows the $\sim S^{-1/2}$ scaling.

both regimes, from ~ 0.03 rad at $S = 10^3$ to $\sim 10^{-3}$ rad at $S = 10^6$ for $U = 0$, confirming that the shot noise is a genuine finite-sampling effect rather than an analytic surrogate. The $U = 10$ curve sits a few times higher at the same S , because its $4\times$ larger parameter count and its energy coefficients of size ~ 10 give a larger per-shot variance on the geometric connections and the dynamical leg. Crucially, this shot spread is a *sub-leading* contribution to the mean deviation: the mean deviation curves are unchanged when the shot count is raised from 10^4 to 3×10^5 (Sec. 3.2), so the threshold is set by the incoherent gate error, while the shot noise only sets the seed spread. The minimum-shots floor—the shot count at which this spread first drops below ~ 0.03 rad—is $S^* \approx 1000$ for $U = 0$ and $S^* \approx 30000$ for $U = 10$.

3.4 Output-state infidelity

The phase deviation is not the only quantity that degrades. Figure 3 shows the output-state infidelity $1 - \langle \Psi_{\text{noiseless}} | \rho_{\text{noisy}} | \Psi_{\text{noiseless}} \rangle$ of the final loop state alongside the phase deviation, as ξ grows. The infidelity rises monotonically and saturates near unity by $\xi \sim 3$ for both $U = 0$ configurations, while the phase deviation saturates near the uniform-random value $\pi/2$ rather than continuing to grow. Notably, the phase deviation for $U = 0$ stays below the tolerance well past the point where the state itself is already $\sim 30\%$ infidel ($\xi \approx 0.3$), reflecting the geometric character of the observable: a partially depolarized state can still carry a usable Berry phase. For $U = 10$ the infidelity and the phase deviation rise together from a much lower ξ , because the deeper circuit accumulates more incoherent error per unit ξ .

4 Mechanism

The central question is whether the noiseless phase-cancellation robustness survives stochastic noise. We decompose φ_B into its three legs and analyze their errors separately.

Deterministic bias partially cancels. The two contributions that co-evolve along the shared frozen trajectory— φ_{qc} and the geometric leg $\varphi_{G,2}$ —inherit a *deterministic* bias from, e.g., a coherent over-rotation or the finite- T non-adiabatic residual whose noiseless cancellation is derived in the original paper’s appendix. Because they are evaluated on the same trajectory, this bias can partially cancel in the sum.

Stochastic fluctuations add. By contrast, the shot and incoherent fluctuations on the separate φ_{qc} , $\langle \hat{\mathcal{H}} \rangle$, and connection circuits are drawn independently; sharing the trajectory does not make them cancel. Their variances add in quadrature, so the total phase error grows roughly as the square root of the number of independently sampled quantities.

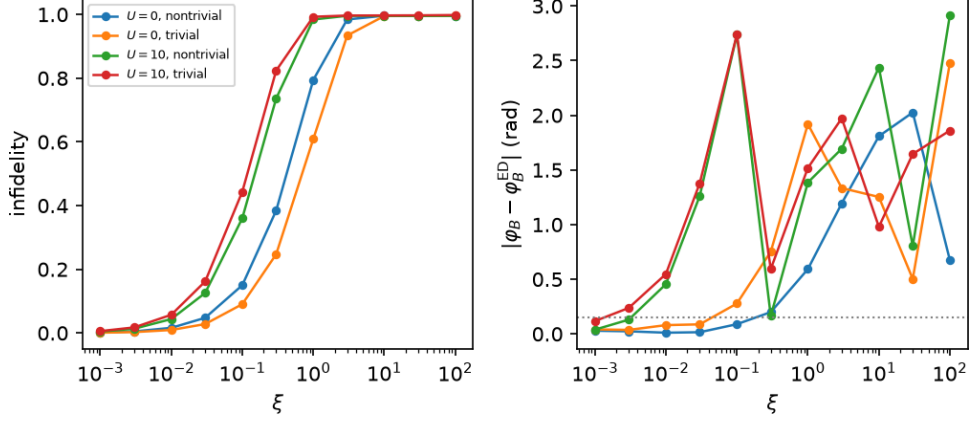


Figure 3: Left: output-state infidelity versus ξ . Right: Berry-phase deviation versus ξ (same data as Fig. 1). The infidelity saturates at unity while the phase deviation saturates near the uniform-random value.

Table 2: Legs of φ_B for $U = 0$ nontrivial (one seed) as ξ grows. $\varphi_{G,1}$ drifts positive and $\varphi_{G,2}$ shrinks, partially cancelling.

ξ	φ_{qc}	$\varphi_{G,1}$	$\varphi_{G,2}$	φ_B
0.03	-1.92	0.46	4.57	3.11
0.10	-1.91	0.54	4.46	3.08
0.30	-1.93	0.75	4.12	2.94
1.00	-1.88	1.10	3.37	2.60

The dynamical leg is a random walk. The leg $\varphi_{G,1} = -\int \langle \hat{\mathcal{H}} \rangle dt$ integrates noisy energy expectations over the loop. Each energy sample is biased toward the noisy value and carries gate and shot noise; integrating over the loop turns this noise into a non-cancelling random-walk contribution, physically distinct from the co-evolving $\varphi_{qc}/\varphi_{G,2}$ overlaps. We therefore decompose φ_G into its $\varphi_{G,1}$ and $\varphi_{G,2}$ legs rather than treating it as one lumped quantity, and support the picture with a per-channel isolation: sweeping each channel with the others off.

What dominates, quantitatively. Table 2 shows the three legs for the $U = 0$ nontrivial case (one seed) as noise grows. The Hadamard readout φ_{qc} is the most robust leg: it measures a single overlap, so its noise is a bounded $\sim 1/\sqrt{S}$ fluctuation. The dynamical leg $\varphi_{G,1}$ drifts monotonically positive (from 0.46 to 1.10 rad as ξ grows to 1): the noisy energy expectation is biased and its integral over the loop no longer cancels exactly. The geometric leg $\varphi_{G,2}$ shrinks (from 4.57 to 3.37 rad): the noisy connection overlaps have reduced magnitude, contracting the accumulated geometric phase. The two drifts have opposite signs and *partially* cancel in $\varphi_B = \varphi_{qc} + \varphi_{G,1} + \varphi_{G,2}$, but only the deterministic (trajectory-shared) part cancels; the stochastic part does not. For $U = 10$ the dominant effect is the *incoherent gate error that the deeper circuit accumulates*, amplified by the method’s own bookkeeping: the energy expectation feeding $\varphi_{G,1}$ carries coefficients of size ~ 10 (the on-site interaction and chemical potential), so a fixed gate error shifts $\varphi_{G,1} \sim 10\times$ more than for $U = 0$, and the geometric leg is multiplied by the large parameter velocities of the over-parameterized interacting ansatz ($|\dot{\theta}_\mu|$ reaching ~ 20). The 3–4 $\times$ deeper circuit and this $\sim 10\times$ energy scale together explain the $\sim 10^2\times$ lower threshold of the interacting regime; the finite-shot spread, by contrast, is sub-leading and is not what limits the threshold (Sec. 3.3).

5 Failure analysis and computational cost

The failure is gradual, not sudden: the trivial/nontrivial discrimination margin closes smoothly as ξ grows, and the seed-to-seed spread widens. At very high noise ($\xi \gtrsim 10$) the estimated phase

becomes effectively uniform on the circle, and the deviation fluctuates around the $\pi/2$ value expected for a uniform phase rather than continuing to grow. The whole-method cost is dominated by the geometric-connection measurements: one forward pass per integration node plus $O(N_\theta)$ connection measurements per node. With $N_\theta \approx 35$ ($U = 0$) or ≈ 143 – 232 ($U = 10$) and ~ 300 – 600 integration nodes, a single (noise level, seed) point costs on the order of minutes for $U = 0$ and a few hours for $U = 10$ on a CPU core, and the sweep is embarrassingly parallel across (noise level, seed, configuration).

6 Related work

The Berry phase was introduced by Berry [1] and underlies the modern classification of topological insulators. Cyclic adiabatic evolution for Berry-phase estimation in gate-based simulation was developed by Murta, Catarina, and Fernández-Rossier [3]. Adaptive variational quantum algorithms—AVQITE [4] and AVQDS [5]—compress ground-state preparation and real-time evolution; the algorithm we characterize combines them as in Mootz and Yao [2]. Noise modeling of variational quantum algorithms and readout is standard (e.g. depolarizing channels, thermal relaxation, and finite-shot statistics); our contribution is the *whole-method* noise-resilience characterization of a topological observable, in which both the readout and the geometric-phase integrands are sampled under noise.

7 Limitations and conclusions

We characterized the noise resilience of adaptive-variational Berry-phase computation on a correlated topological model. The method remains accurate below a device-referenced noise threshold that depends on circuit depth and on the energy scale of the model, and the mechanism is a competition between the partially-cancelling deterministic bias of the co-evolving phase legs and the add-in-quadrature stochastic fluctuations, with the dynamical leg contributing a random-walk drift and the geometric leg amplifying the incoherent connection noise through the large parameter velocities of the over-parameterized ansatz. The interacting regime is $\sim 10^2 \times$ more fragile than the noninteracting one because its deeper circuit and larger energy scale amplify the gate error; the finite-shot spread is a sub-leading, separately reported effect. The main limitation is scope: the ansatz and trajectory are frozen at noiseless values, so the threshold characterizes the resilience of the measured phase at a fixed trajectory, not the noise-driven re-growth of the ansatz (a harder study we leave to future work). A Trotterized baseline at matched noiseless accuracy would sharpen the shallow-vs-deep comparison.

References

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